Urban Pulse - Tourism & Culture Intelligence Platform

Sensing India's seasonal pulse: from monument footfall to festival demand

MILESTONE 1: DOMAIN FINALIZATION & FOUNDATION

TEAM - 1

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Problem Statement

India's heritage sites and festivals form the cultural "pulse" of its cities - but this pulse is sensed only **reactively**. Monuments get overcrowded during festivals with no advance warning. Accessibility for senior citizens, disabled visitors, and low-income domestic tourists is rarely measured. Weather disruptions aren't factored into visitor planning. Tourism data stays siloed across ASI, state tourism boards, and the Ministry of Tourism.

Urban Pulse - Tourism & Culture aims to change that: a smart, data-driven platform that senses the real-time and seasonal health of India's tourism ecosystem, turning fragmented government data into actionable insight.

Problem Objectives

1. **Real-Time Footfall Tracking**
   Track real-time monument footfall for both domestic and foreign visitors across India's heritage sites.
2. **Seasonal Demand Sensing**
   Sense seasonal and festival-driven demand patterns to anticipate surges before they happen.
3. **Site Utilization Analysis**
   Identify overcrowded vs. underutilized heritage sites to enable smarter resource distribution.
4. **Economic Impact Quantification**
   Quantify economic impact across states to support evidence-based tourism policy.
5. **Footfall Forecasting**
   Enable forecasting of future footfall by season and event calendar for proactive planning.

Project Scope

In Scope

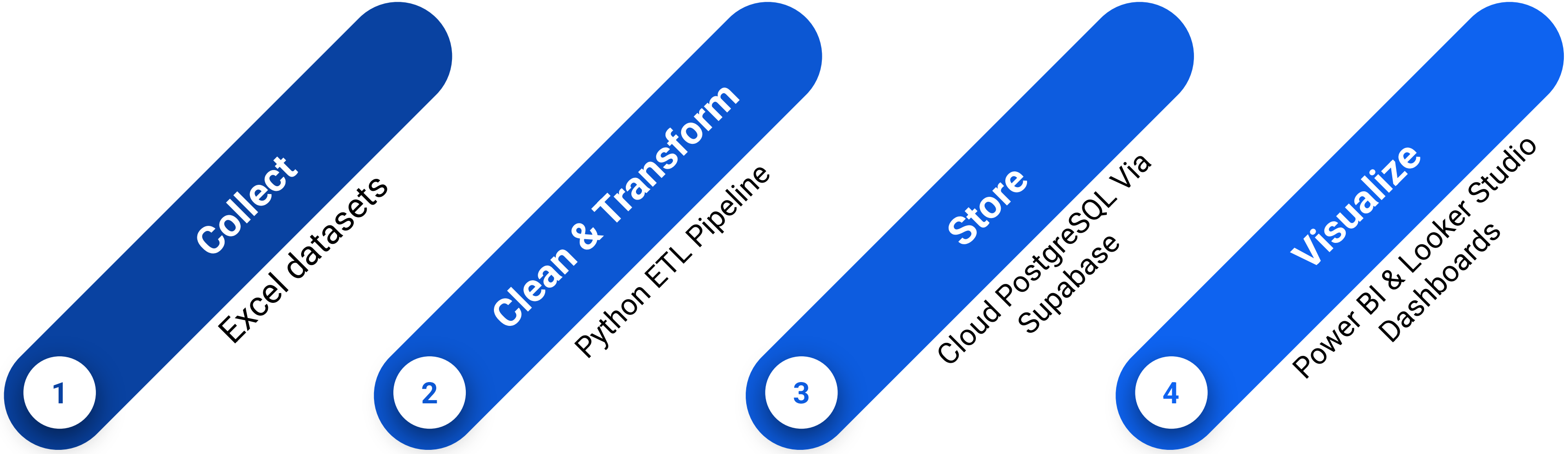
- India-specific monument visitor data (domestic + foreign)
- Festival and cultural mela participation & funding, state-wise
- Seasonal and festival driven demand patterns
- Accessibility indicators for vulnerable visitor groups
- Galaxy schema data model (4 fact + 3 shared dimension tables)
- Interactive dashboards - geospatial, trends, executive views

Out of Scope

- Global/cross-country tourism comparisons
- Outbound travel (Indians traveling abroad)
- Everyday commuter mobility (covered by the Mobility team)
- Hotel occupancy and booking systems
- Real-time IoT sensor deployment

Proposed Solution

A unified data intelligence platform connecting monument footfall and festival participation through a shared Galaxy schema, visualized in interactive dashboards.



Each step in the pipeline is designed to be reproducible, cross-platform, and aligned with verified Government of India data sources - ensuring the platform is genuinely deployable, not just a prototype.

Technology Stack

Python (pandas)

ETL: extracting, cleaning, and transforming raw datasets into structured form.

Supabase

Free cloud PostgreSQL database - shared and cross-platform across Mac and Windows.

Power BI & Looker Studio and dbdiagram.io

Dashboard visualization and executive reporting for stakeholders & policymakers. Looker Studio is deployed as web based dashboard solution for seamless cross platform access on macOS

GitHub

Version control for scripts, schema docs, and team collaboration across the project lifecycle.

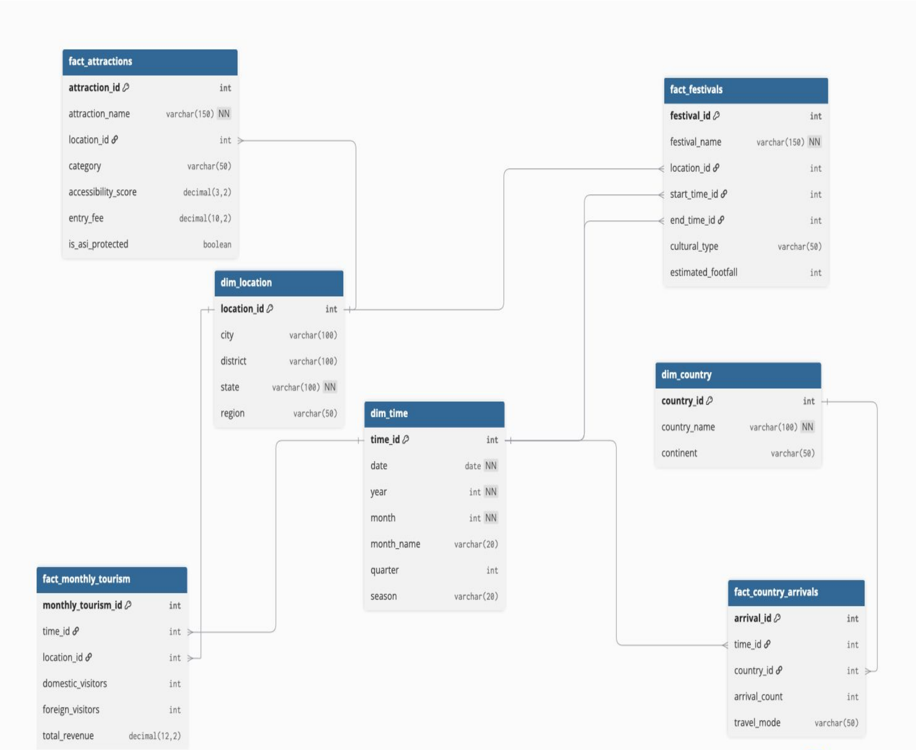
Supabase was chosen specifically because our team works across both Mac and Windows environments.

About Our Datasets

All data is sourced from verified Government of India portals and official ministry reports.

Source	Coverage	Used For
Ministry of Tourism / IITTM Report	Tourist places, entry fees	Fact_attractions
data.gov.in (DPPH Scheme & NE)	State-wise festival funding (2014–2024)	Fact_festivals
data.tourism.gov.in	Monthly revenue & foreign arrivals (2019–2022)	fact_monthly_tourism
data.tourism.gov.in	FTAs by country, age demographics, avg stay duration	fact_country_arrivals
Ministry of tourism/ASI	Master lookups for Time, Cities/States, and Foreign Countries	Dim_time, dim_location, dim_country

Methodology



Database ER diagram modeled via dbdiagram.io.

Galaxy Schema *(Fact Constellation)*

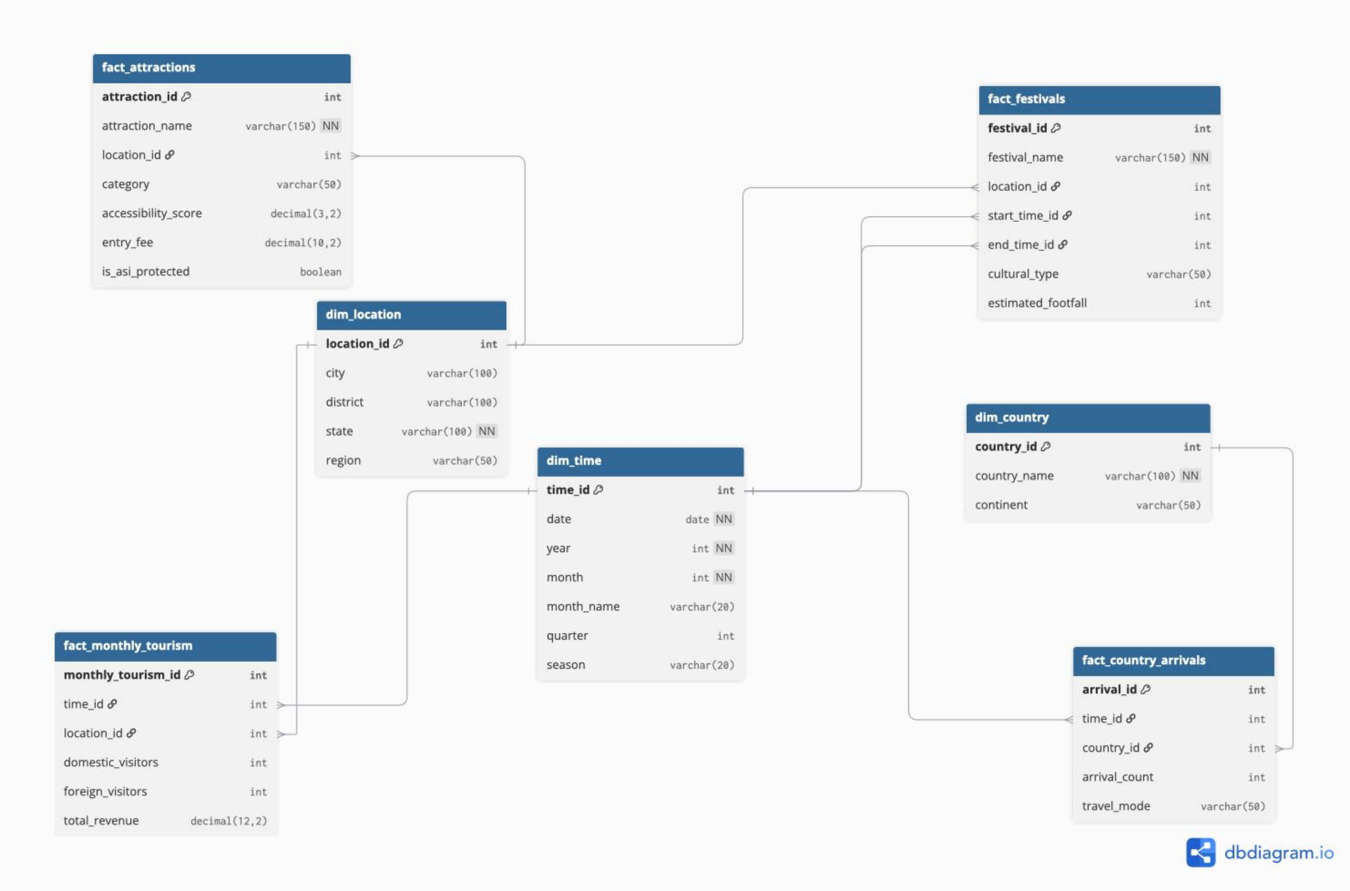
We use a **Galaxy Schema** consisting of **4 Fact tables sharing 3 core Dimension tables**:

- **Dimension Tables (Shared Context):** `dim_time`, `dim_location`, and `dim_country`.
- **Fact Tables (Specific Metrics & Granularities):** `fact_attractions`, `fact_festivals`, `fact_monthly_tourism`, and `fact_country_arrivals`.

> We chose a Galaxy Schema because our domain spans multiple events at different levels of detail (point-in-time spot ratings vs. monthly revenue vs. demographic percentages). Structuring it this way prevents data duplication and keeps Power BI measures fast and accurate.

Progress Snapshot

Database schema (4 fact, shared dimensions) modeled in dbdiagram.io and deployed on Supabase PostgreSQL



TOURISM & CULTURAL ANALYSIS

TOURISM & CULTURAL ANALYSIS DASHBOARD

159M	327K	1000	279	4.46	102.84
Total Tourist Arrivals	Total Tourism Revenue	Total Attractions	Total Festivals	Attraction Rating	Entry Fee

FILTERS

Year
All

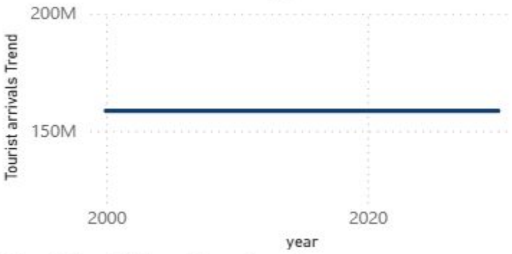
State
All

City
All

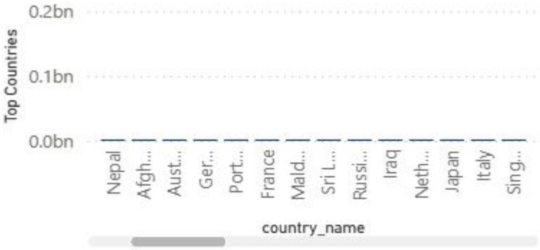
Country
All

Category
All

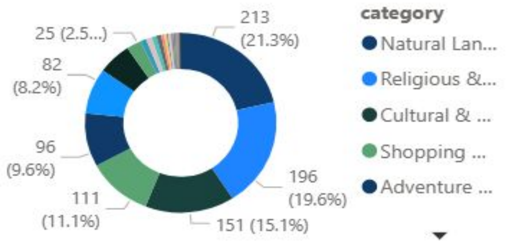
Tourist arrivals Trend by year



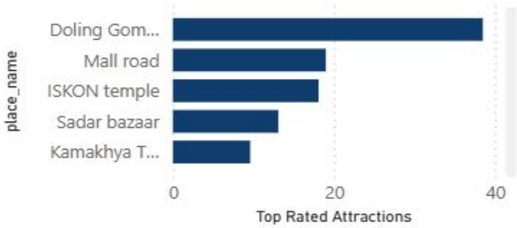
Top Countries by country_name



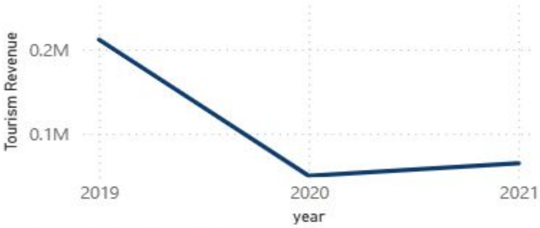
Attractions by category



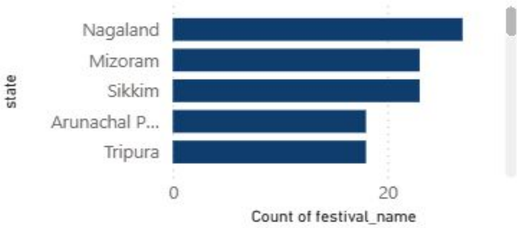
Top Rated Attractions by place_name



Tourism Revenue by year



Count of festival_name by state



country_name	Sum of arrivals_in_numbers
Afghanistan	36451
Argentina	530
Australia	33864
Austria	4411
Bahrain	1727
Bangladesh	240554
Belgium	7382
Total	**158546874**

city and place_name

Sum of amount_sanctioned and Sum of amount_released by festival_name

Expected Outcomes

→ Unified Dashboard

A single dashboard covering both monument footfall and festival participation - giving stakeholders a complete picture of India's cultural tourism ecosystem.

→ Seasonal Forecasting

Predictive models anticipating demand spikes by festival, season, and weather - enabling proactive planning rather than reactive crisis management.

→ Accessibility Insights

Highlighting gaps for seniors, disabled visitors, and low-income tourists - groups whose needs are rarely captured in existing tourism data.

→ Deployable Policy Model

A genuinely deployable model aligned with **Dekho Apna Desh** -and **Swadesh Darshan** - ready for adoption by state tourism boards and policymakers.

Conclusion

Tourism and Culture together form the **living heritage layer** of India's Urban Pulse - distinct from everyday mobility, yet built on the same core philosophy: sensing a city's pulse and turning fragmented data into actionable insight.

Verified Data

Grounded in verified Government of India data from ASI, Ministry of Tourism, and Ministry of Culture.

Galaxy Schema

Galaxy schema connects monument visits and festival participation through shared dimension

Free & Cross-Platform

Built on free, cross-platform tools - Supabase and Power BI - accessible to the entire team.

Deployable Use Case

A genuinely deployable use case for state tourism boards and policymakers across India.